Tax Shocks and County-Level Migration: Do High-Income Households Flee Higher Taxes?

Ben Hobbs & Bergen Miller

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Contents

1 Introduction

We examine whether changes in state-level tax policy encourage high-income households to relocate to lower-tax jurisdictions.

Our primary data source is the IRS Statistics of Income (SOI) series, which publishes county-level income and migration data annually. The income data reported are adjusted gross income (AGI) and taxes paid by households aggregated to the level of county-income bracket-year. The migration data report the number of returns, number of individuals, and cumulative AGI for every county-to-county move in each year. Our sample spans 2011–2022.

The unit of observation in our analysis is a county-pair $\times$ year. So for each origin county, we observe how many people moved to each destination county from a given year to the next, along with the income and tax characteristics of both counties.

To identify causal effects we will explore three state-level policy shocks with known effective dates:

1. **New Jersey (2018):** top marginal rate raised from 8.97% to 10.75% on income above $5M; threshold later lowered to $1M in 2021.
2. **New York State (2021):** top marginal rate raised from 6.85% to 9.85% on income above $1M.
3. **Delaware (2018):** estate tax repealed (previously 16% on inheritances above $5.49M).

Our effective tax rate variable is computed as $(\textit{Federal income tax} + \textit{State}\ \&\ \textit{local taxes paid})/AGI$, computed separately for high-income (AGI $>$ $200k) and low-income (AGI $\leq$ $200k) households in each county-year.

2 Data Construction

All the data we use is scraped from the IRS Statistics of Income (SOI) webpage. The two data series we use are county-level aggregates of household income and tax statistics during the period of 2011-2022. The IRS bins households by adjusted gross income (AGI), so information about households with AGI above $200k is separate from those with AGI below $200k (we combine all the other AGI bins and call these two groups high and low-income households). To protect privacy, the counties where the number of tax returns in an AGI bin is fewer than 20 have all the values for that county-year-AGI bin combination suppressed to zero.

Migration data comes from a separate series of publications also from SOI. It includes the number of tax returns which were filed in one county in year 1 and another in year 2, the fips codes of those two counties, the number of people represented in the tax returns, and the total AGI of these tax returns in the second year. We construct a panel of all the county migration pair-year combinations with various income and tax statistics about the two counties. Similar to the income and tax data, if the number of tax returns in a county pair-year combination is fewer than 10, the values for that observation are suppressed to -1.

Our analysis is not so concerned with the extensive margin (a county having any migrants), so we drop all observations where migration data are suppressed to -1. These observations are entirely very small counties, and we expect this decision not to bias any estimates. In the case where a county's income data is suppressed, the effective tax rates are NaN, and those counties are omitted from regressions.

3 Descriptive Analysis

The figures below explore the unconditional relationships in our data before turning to causal identification. We ask two preliminary questions:

1. Across all county-pairs, do wealthier migrants tend to move toward lower-tax destinations?

2. Does this pattern differ depending on whether we measure the tax rate faced by high-income or low-income households?

3.1 Migrants' Wealth vs. High-Income Tax Differential

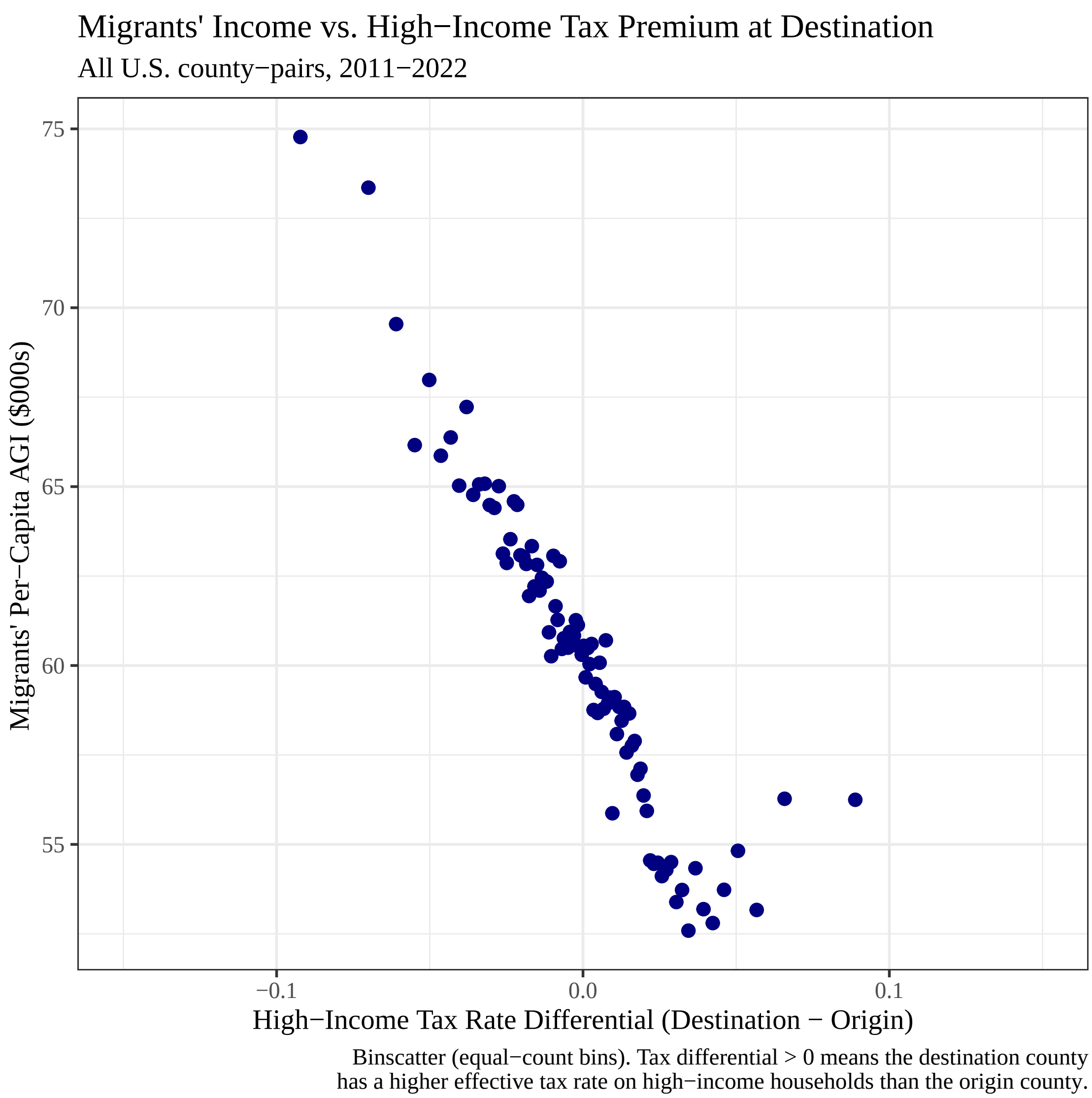


Figure 1: Migrants' income vs. high-income tax premium at destination.

The downward slope of the figure indicates that the type of counties into which migrants have higher average incomes are counties where households with AGI above $200k happen to pay lower tax rates than the county those migrants left. This is consistent with the basic logic of 'rich people evading taxes.'

3.2 Migrants' Wealth vs. Low-Income Tax Differential

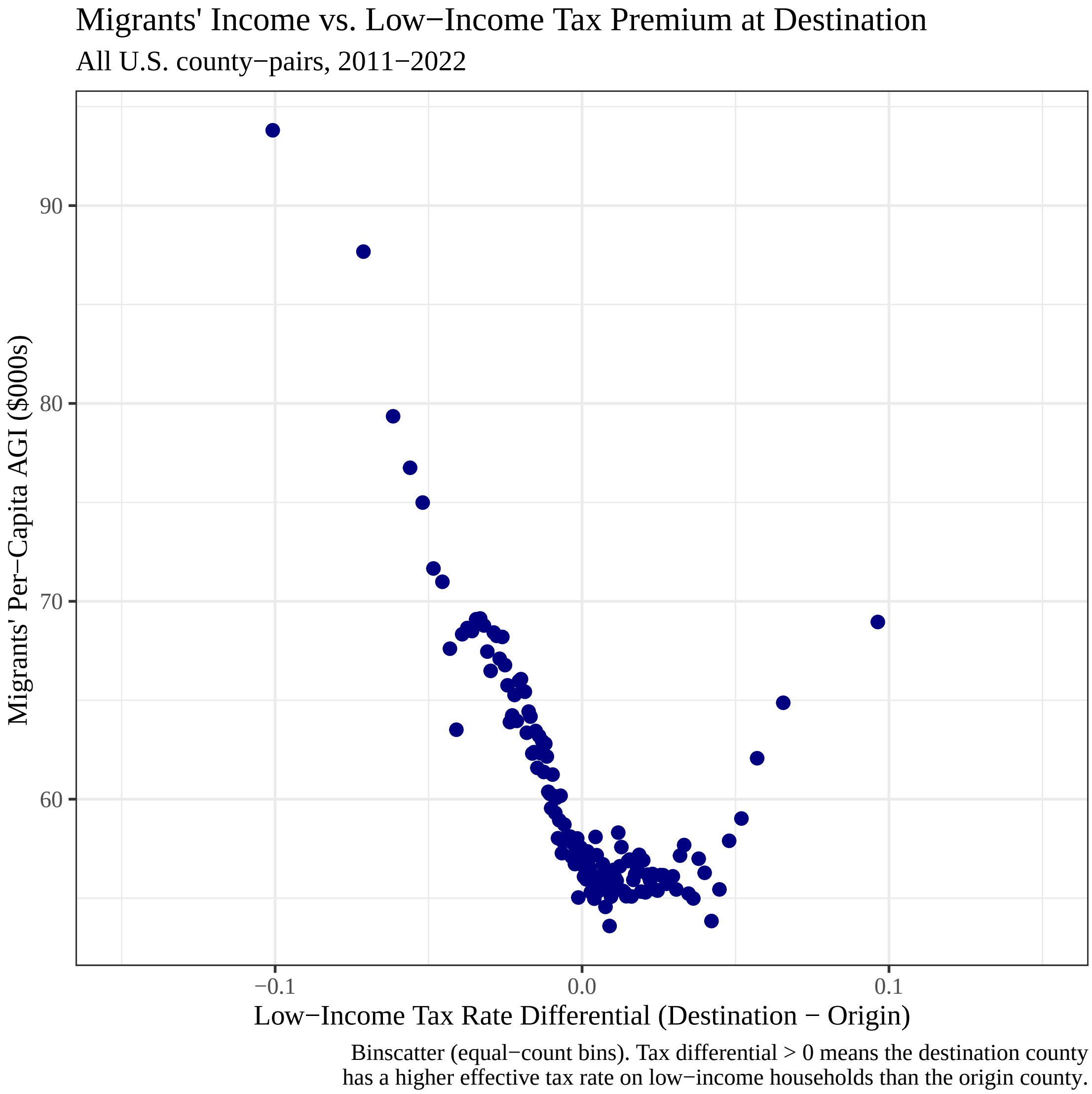


Figure 2: Migrants' income vs. low-income tax premium at destination.

The pattern here is similar to Figure 1. Migrants' income is still negatively correlated with higher relative realized taxes in the destination county. This suggests positive correlation between taxes faced by high-income households and low-income (the bottom ~95%) households. We do not know how to account for the curve upwards near tax-premium = 5 pp.

4 Manhattan Case Study

To reduce cross-sectional noise we will look Manhattan (New York County, FIPS 36061), one of the highest-tax, highest-income counties in the country. We ask: among the people leaving Manhattan NYC in any given year, does the destination county's effective high-income tax rate influence those migrants' average incomes?

A negative slope would suggest high-earning Manhattanites prefer moving to lower-tax destinations. A positive slope, which is what we observe, suggests something more nuanced: Manhattan-leavers tend to relocate to other high-tax metros (San Francisco, Boston, Seattle), rather than moving to minimize taxes.

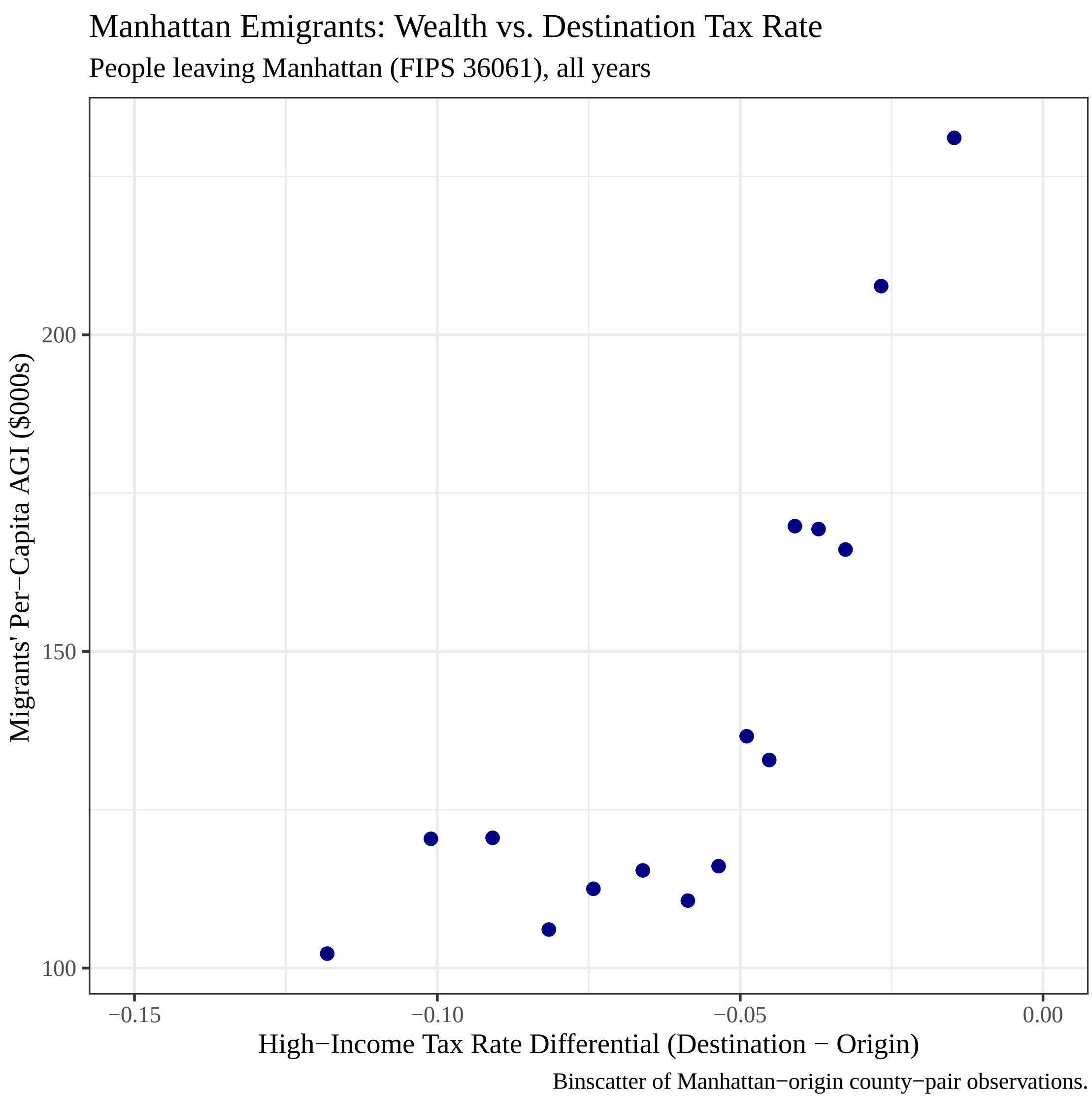


Figure 3: Manhattan emigrants' wealth vs. destination tax rate.

Across every year in our data, figure 3 bins observations by the size of the difference between high-income households' taxes in the destination county and Manhattan County in NYC. Interesting, the bins are such that every measurement of the Conditional Expectation Function has a negative tax differential value (which makes sense, because NYC is a high-tax place). Where the tax discount in the destination is less than 5 pp, there appears to be no relationship with the income of the people moving there. Counter-intuitively, and counter to the implications of figures 1 & 2, counties with higher realized taxes on high-income households (similar to NYC) attracted migrants with higher average incomes. This is explained by the allure of places like San Francisco outweighing the dreaded California tax code.

Something to notice about figure 3 is that every bin has a very high average income. Peering into the data some more, there are indeed some grotesquely rich people, featuring 166 households who filed their taxes in Manhattan in 2012 and in Albany county in 2013 who had an average income of $3,800,000.

From 2020 to 2021, there were 2500 households with an average income of $1.2M moving from Manhattan to Miami-Dade County. Not even the Central Limit Theorem can hide the unbelievable right tail of the American income distribution.

5 Regression Analysis

We now turn to the difference-in-differences analysis for each policy in turn.

5.1 New York 2021 Tax Hike

In 2021, New York State raised its top marginal income tax rate from 6.85% to 9.85% on income exceeding $1 million. Our strategy capitalizes on the fact that some destination counties outside New York were already attracting rich New York migrants before the shock, while others were not. If the 2021 tax increase caused a behavioral response among high earners, the counties that were drawing wealthier New Yorkers in the pre-shock period should see a larger increase in inflows after the rate rose. If these counties also especially attracted the ultra-high-income households, these migrants' average incomes should increase too.

The figure below plots the proportional change in migration flow (average of 2021-2022 moves vs. the 2019 pre-shock baseline) against the log per-capita AGI of people who moved from each NY county to each destination in the pre-shock year. A positive slope would support the hypothesis that high-income New Yorkers were pushed out at an increased rate after the tax increase.

> **Note on year labels:** Each `year` value in the data is the *starting* year of the IRS migration file, so `year == 2019` captures people who physically moved between filing their 2019 taxes and 2020 taxes, `year == 2020` captures moves between 2020 taxes and 2021 taxes (the immediate shock year), and `year == 2021` captures moves during 2022. The pre-shock baseline uses `year == 2019` (moves in 2020); the post-shock period averages `year %in% c(2020, 2021)` (moves in 2021 and 2022).

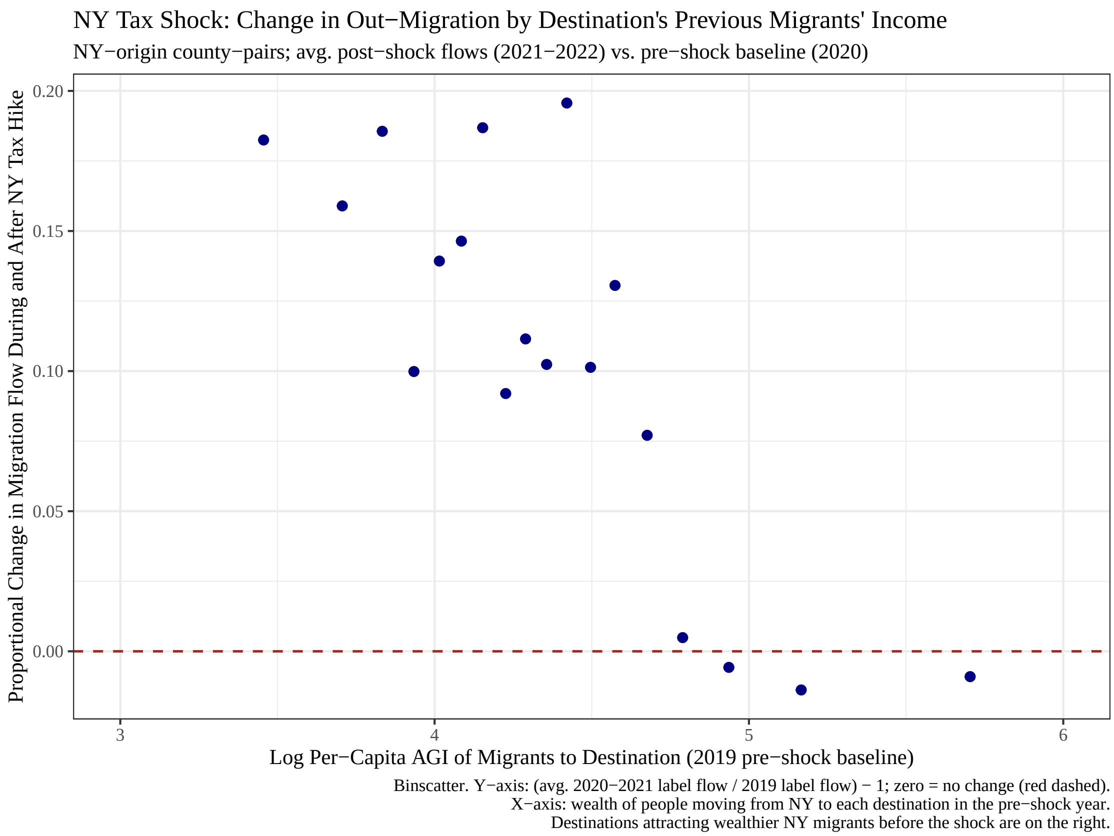


Figure 4: New York tax shock: change in out-migration by destination's previous migrants' income.

The binscatter suggests a negative relationship between pre-shock migrant income and the subsequent increase in out-migration from New York. Among previously low-income migration channels, the relationship is noisy, but the percent increase after the NY tax hike in county migration is sharply negative for counties whose previous recipients from NY had an average income of more than $116k.

The map below shows the *geographic* version of the same question: which parts of the country absorbed New York migrants, and how did that pattern shift in the year taxes went up?

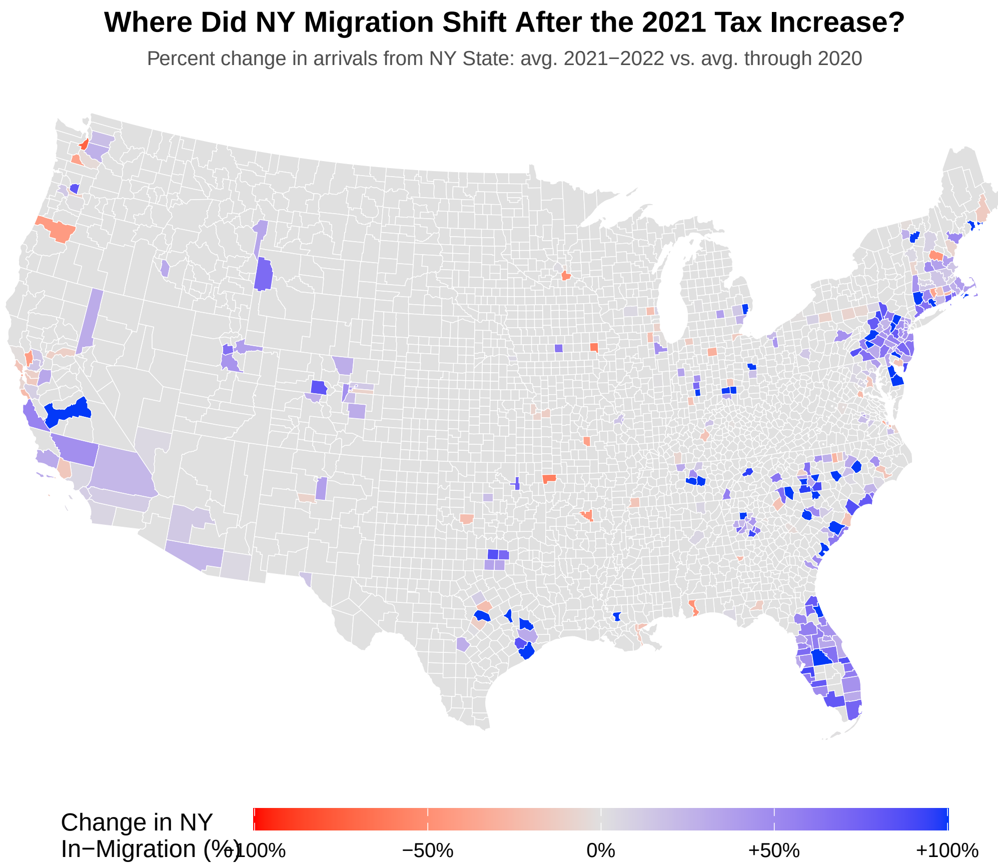


Figure 5: Geographic redistribution of New York out-migration before vs. after the 2021 tax increase.

The map complements the binscatter results. Blue counties absorbed more New York migrants after the 2021 rate increase; red counties absorbed fewer. The gains are concentrated in Florida (particularly Miami-Dade and Palm Beach counties), parts of the Carolina and Texas metros, and a handful of coastal destinations. Most of the country shows little change, meaning the shock did not produce a broad nationwide redistribution of New York migrants. The pattern is geographically clustered rather than uniformly driven by low taxes, pointing toward amenity sorting as a competing explanation alongside tax avoidance.

> **Note:** An animated version of this map showing Manhattan out-migration year-by-year from 2011–2021 is available in `Updated Working Files/Analysis/Map Visualization.R`. Run that script to produce `manhattan_exodus_animated.gif`. The animation makes the 2021 shift visually apparent against the prior trend.

> **Disclaimer:** The policy change happened at the same time as the pandemic. It is impossible to distinguish between the effect of the tax change and the effect of the pandemic in New York City, so any causal interpretation of the New York estimates is seriously confounded.

To estimate how the tax change affected out-migration from New York in the full county-pair panel, we begin with the following equation:

$$log(n) = \alpha + \beta_1(Treat_c) + \beta_2(Post_t) + \beta_3(Treat * Post)_{c,t} + \epsilon_{c,t}$$

where n is the number of tax returns moving from one county to another, $Treat_c$ is an indicator for if the county in year 1 was in New York, and $Post_t$ is an indicator for if the year of the observation was 2020 or 2021 (to capture people who moved anticipating the increase in taxes as well as people responding to the increase by leaving the following year). We also estimate the income composition model, where the only difference is that number of returns has been replaced by mean AGI of those tax returns:

$$log(\frac{AGI}{n}) = \alpha + \beta_1(Treat_c) + \beta_2(Post_t) + \beta_3(Treat * Post)_{c,t} + \epsilon_{c,t}$$

For all income specifications and income-trend plots, we drop county-pair observations with non-positive migrant per-capita AGI before taking logs. These are rare rows where the IRS migration file reports zero or negative aggregate AGI, and leaving them in would create artificial $-\infty$ values and misleading dips in the plotted averages.

Table 1: NY 2021 DiD Estimates. Standard errors in parentheses.
*** $p < 0.001$.

Coefficient	Volume: $\log(n)$	Income: $\log(AGI/n)$
β_1 (Treat)	-0.0089 (0.0075)	0.1871*** (0.0036)
β_2 (Post)	0.2366*** (0.0035)	0.2568*** (0.0017)
β_3 (Treat $\times$ Post)	0.0677*** (0.0176)	0.0693*** (0.0085)

There was a 23.7% increase in migration among counties outside of New York from the 2011-2020 average to the 2020-2022 average. The increase between the two periods for people leaving New York was 6.8 pp greater than the national average. As suggested by figure 1, the New York State tax increase did increase the number of New Yorkers fleeing the state.

Among observations with positive migrant per-capita AGI, the average income of people moving out of New York State during 2011-2022 was 19% higher than the people leaving anywhere else. There was a 26% increase in mean AGI of people leaving a county outside of NY from the 2011-2020 average to the 2020-2022 average. From the coefficient estimate of the interaction term, the increase in mean AGI of people leaving NY from the 2011-2020 average to the 2020-2022 average was 7% greater than the mean AGI of people leaving anywhere else. This is consistent with higher-income New Yorkers leaving at a faster rate after the tax hike, though the pandemic makes a clean causal interpretation difficult.

The figures below show mean log migration volume and mean log per-capita AGI over time for New York counties (treated) versus all other origin counties (control). The dashed line marks the start of the post period.

> **Note:** The sharp level jump between calendar 2013 and 2014 in the volume trends is best understood as a reporting break in the IRS county-to-county migration files, not as a sudden nationwide surge in out-migration. The plot compares the IRS `2012-2013` file to the `2013-2014` file, and the IRS's `2013-2014 Migration Data Users Guide` states that the county-file disclosure threshold was raised from 10 returns to 20 returns, with additional county-level suppression. In our cleaned data, nearly 47% of county-pair observations in calendar 2012-2013 had only 10-19 returns, while those small flows are almost entirely absent from calendar 2014 onward. Mechanically removing many small flows raises the average of $\log(n)$, so the post-2013 level shift should not be interpreted as an economic break in migration behavior.

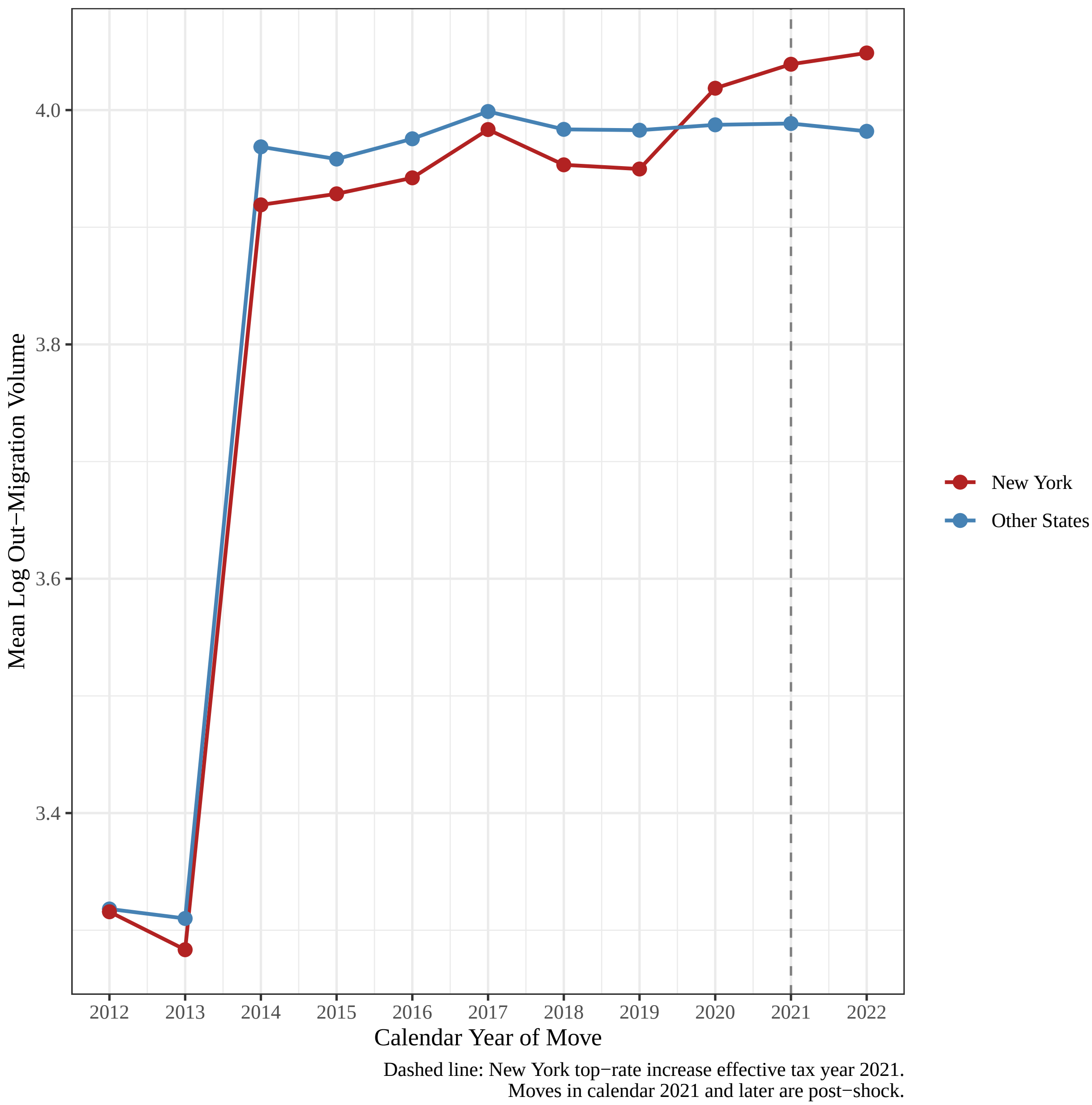


Figure 6: Mean log out-migration volume over time for New York counties vs. other origin states.

The volume trends plot shows that New York out-migration moved broadly with the control group pre-tax increase, but rose sharply after it came into effect. This is consistent with the positive treatment by post coefficient.

Note: A small number of county-pair observations in the IRS migration files report zero or negative total AGI even when the number of returns is positive. Because the income trends and income regressions use $\log(AGI/n)$, keeping those rows would mechanically create artificial $-\infty$ values and misleading dips in the plotted averages. We therefore drop observations with non-positive migrant per-capita AGI before taking logs in the income specifications only. This is appropriate here because the goal of these plots is to compare the intensive-margin income composition of observed migration flows, not to treat zero or negative reported AGI as economically meaningful average migrant income.

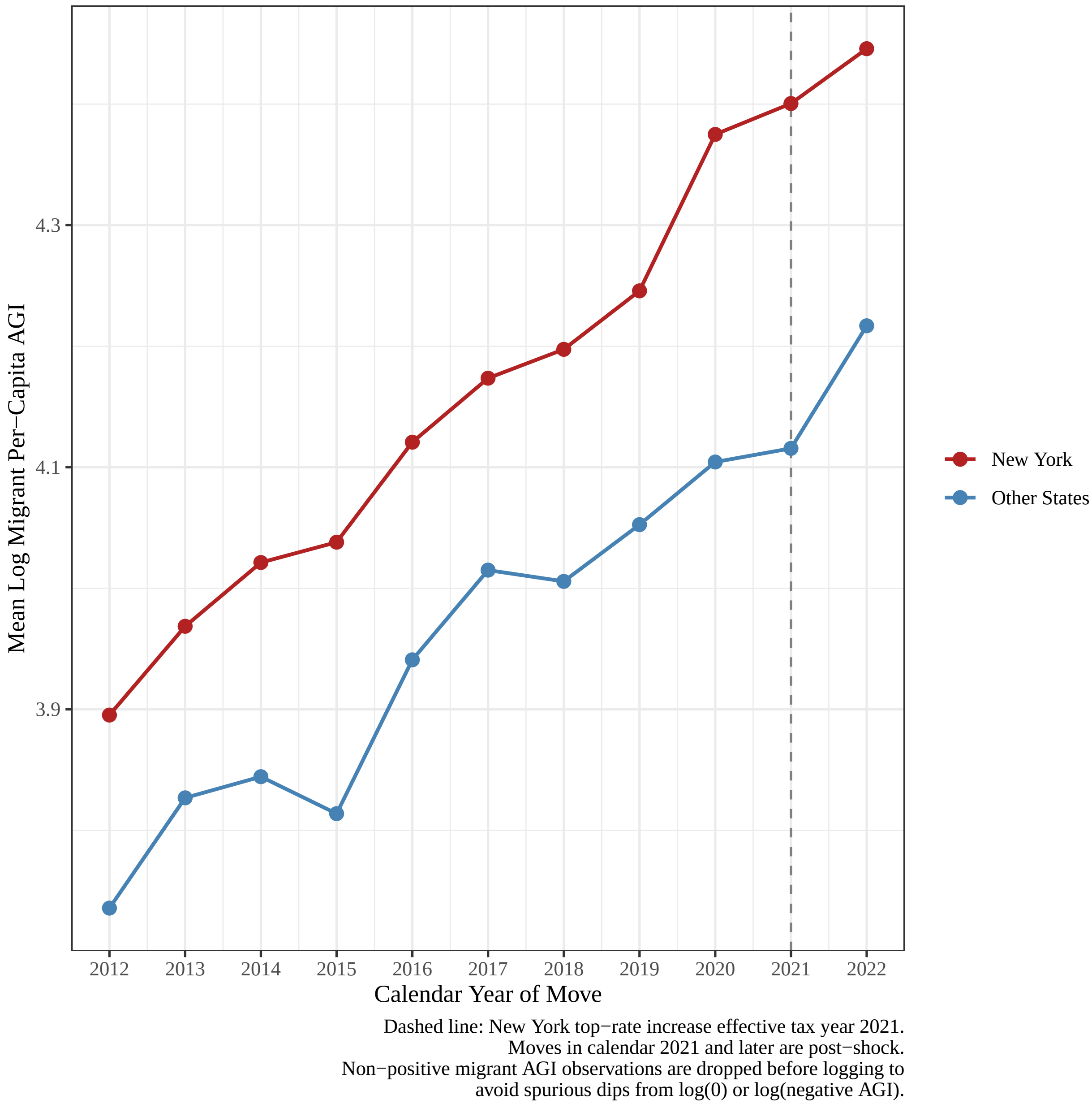


Figure 7: Mean log migrant per-capita AGI over time for New York counties vs. other origin states.

The income trends plot shows that New York out-migrants had higher average per-capita AGI than migrants from other states throughout the period. Post 2021, the New York-control gap still appears to widen somewhat, which is consistent with the positive interaction coefficient in table 1.

5.2 New Jersey 2018 Tax Hike

New Jersey raised its top marginal income tax rate in 2018, increasing the top rate from 8.97% to 10.75% for the highest income households. Because this policy change occurred well before the pandemic and leaves us with more post-event data, it provides a useful comparison case for the New York analysis. For this regression, the treatment is county-pair observations where the county of origin is in New Jersey.

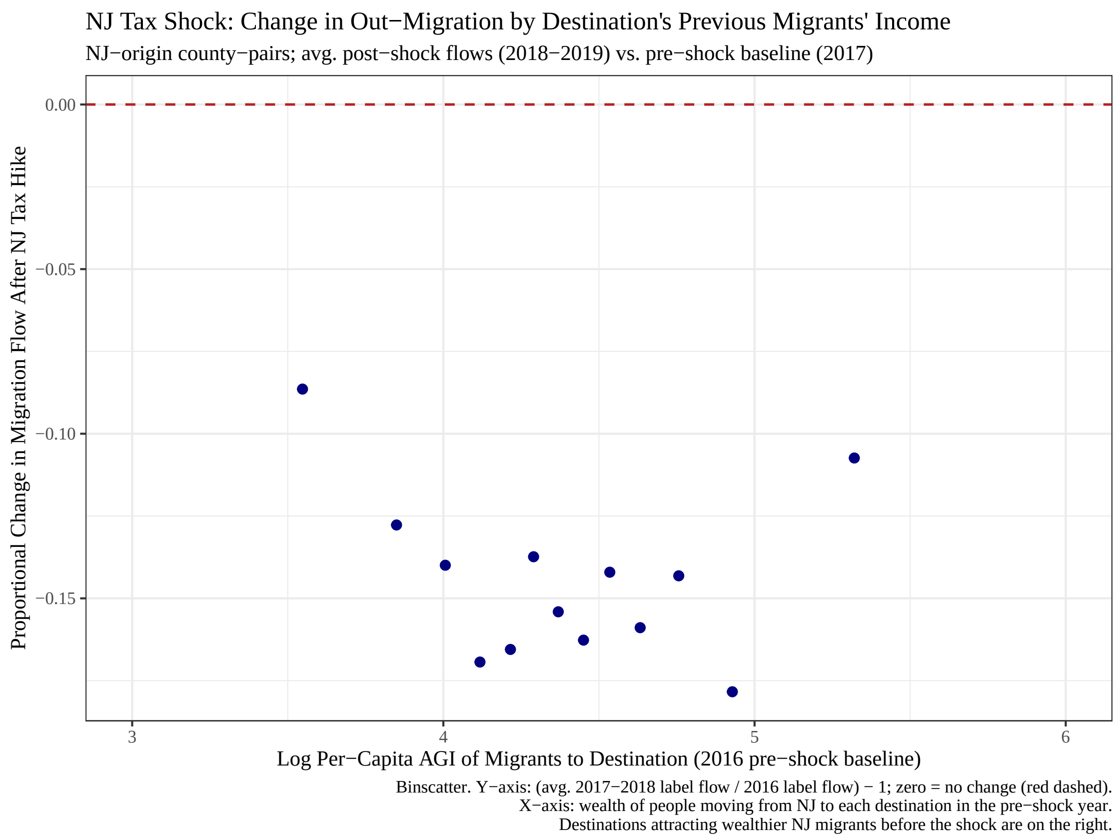


Figure 8: New Jersey tax shock: change in out-migration by destination's previous migrants' income.

The binscatter above suggests no relationship between pre-policy-change migrant incomes and the subsequent change in the number of households leaving New Jersey. Rather, everybody was less likely to leave NJ after the policy change.

We estimate the effect on out-migration volume:

$$log(n) = \alpha + \beta_1(Treat_c) + \beta_2(Post_t) + \beta_3(Treat \times Post)_{c,t} + \epsilon_{c,t}$$

where $Treat_c = 1$ if the origin county is in New Jersey and $Post_t = 1$ for years 2017–2018 (moves during 2018–2019). We also made a model that looks at the income composition of movers:

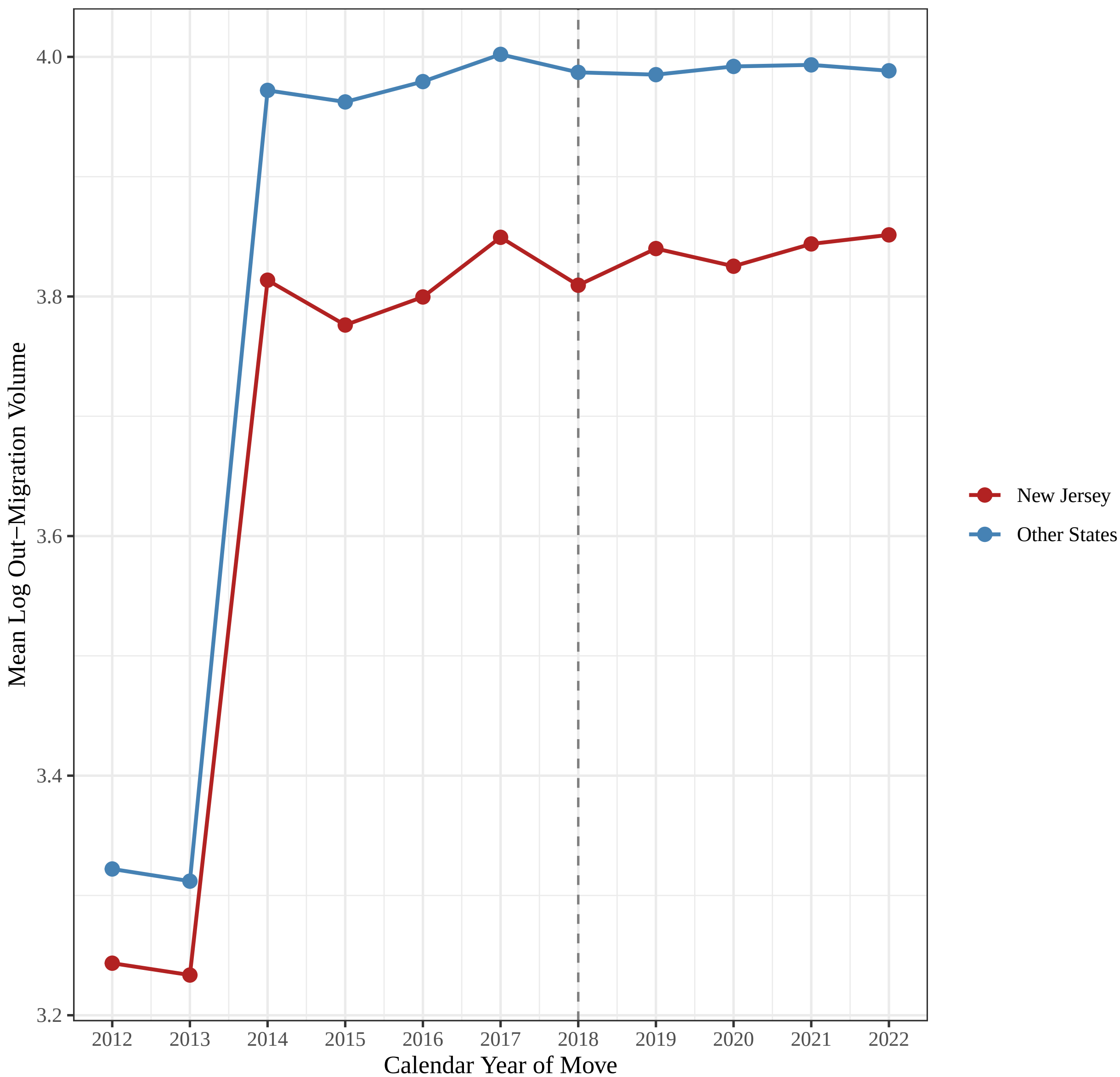
$$log\left(\frac{AGI}{n}\right) = \alpha + \beta_1(Treat_c) + \beta_2(Post_t) + \beta_3(Treat \times Post)_{c,t} + \epsilon_{c,t}$$

Table 2: NJ 2018 DiD Estimates. Standard errors in parentheses.

Coefficient	Volume: $\log(n)$	Income: $\log(AGI/n)$
β_1 (Treat)	-0.1221*** (0.0093)	0.4037*** (0.0045)
β_2 (Post)	0.2281*** (0.0036)	0.0876*** (0.0017)
β_3 (Treat $\times$ Post)	-0.0398. (0.0233)	-0.0038 (0.0114)

The New Jersey estimates do not show strong evidence of a large response to the 2018 tax increase. The volume model shows that tax filer flows that originated in New Jersey were lower than flows from other points of origin on average, and the post-period coefficient indicates that migration volume increased in general after 2017 among the control group. However, the interaction coefficient is negative and only slightly significant, which suggests New Jersey out-migration was slightly lower than other states after the tax hike. In the income model, which again excludes non-positive migrant per-capita AGI before logging, migrants leaving New Jersey had substantially higher mean AGI than migrants leaving other states (40% higher), but the treatment by post coefficient is very small and statistically insignificant. Altogether, this means that the tax increase did not clearly change the income composition of people leaving New Jersey.

The figures below show mean log migration volume and mean log per-capita AGI over time for New Jersey counties (treated) versus all other origin counties (control). The dashed line marks the start of the post period.



Dashed line: New Jersey top−rate increase effective tax year 2018.
Moves in calendar 2018 and later are post−shock.

Figure 9: Mean log out-migration volume over time for New Jersey counties vs. other origin states.

The volume trends plot shows that New Jersey out-migration moves broadly with the control group, and no clear post 2017 surge that would indicate that the change in tax policy resulted in a state-wide exodus.

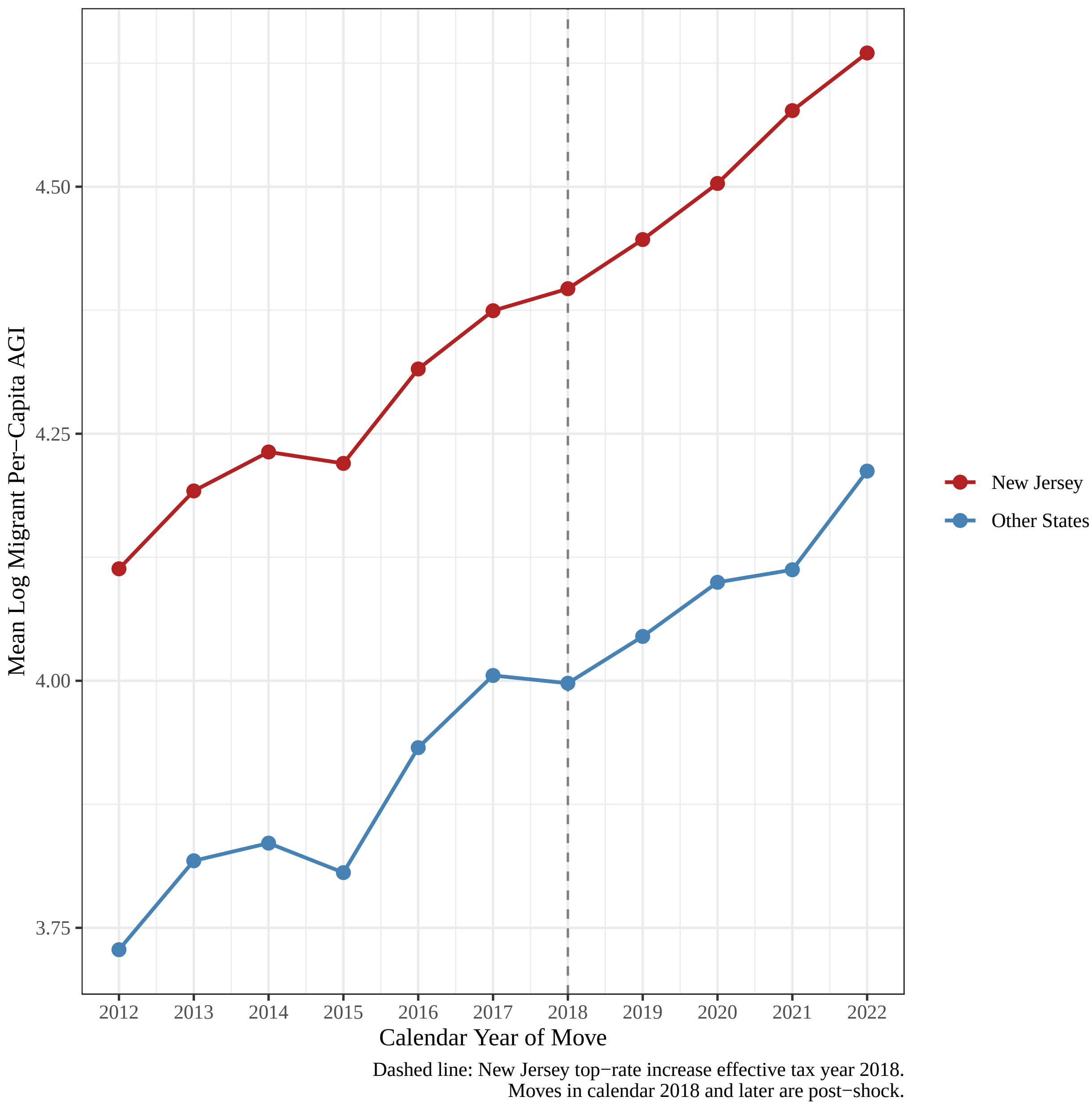


Figure 10: Mean log migrant per-capita AGI over time for New Jersey counties vs. other origin states.

The income trends plot shows that New Jersey out-migrants had higher average per-capita AGI than migrants from other states. However, there is not a clear break after the 2017 tax policy change that would suggest a change in the income composition of movers.

5.3 Delaware 2018 Estate Tax Repeal

Delaware repealed its estate tax in 2018, which was previously only pertinent to very large estates. Compared to the previous analyses, this policy is a tax cut rather than an increase, so the expected behavior would be in the opposite direction. Instead of looking at whether individuals are leaving a high tax state, were looking at whether the repeal made Delaware more attractive (on top of other effects). Because of this, we decided to put the treatment on the destinations side rather than origin.

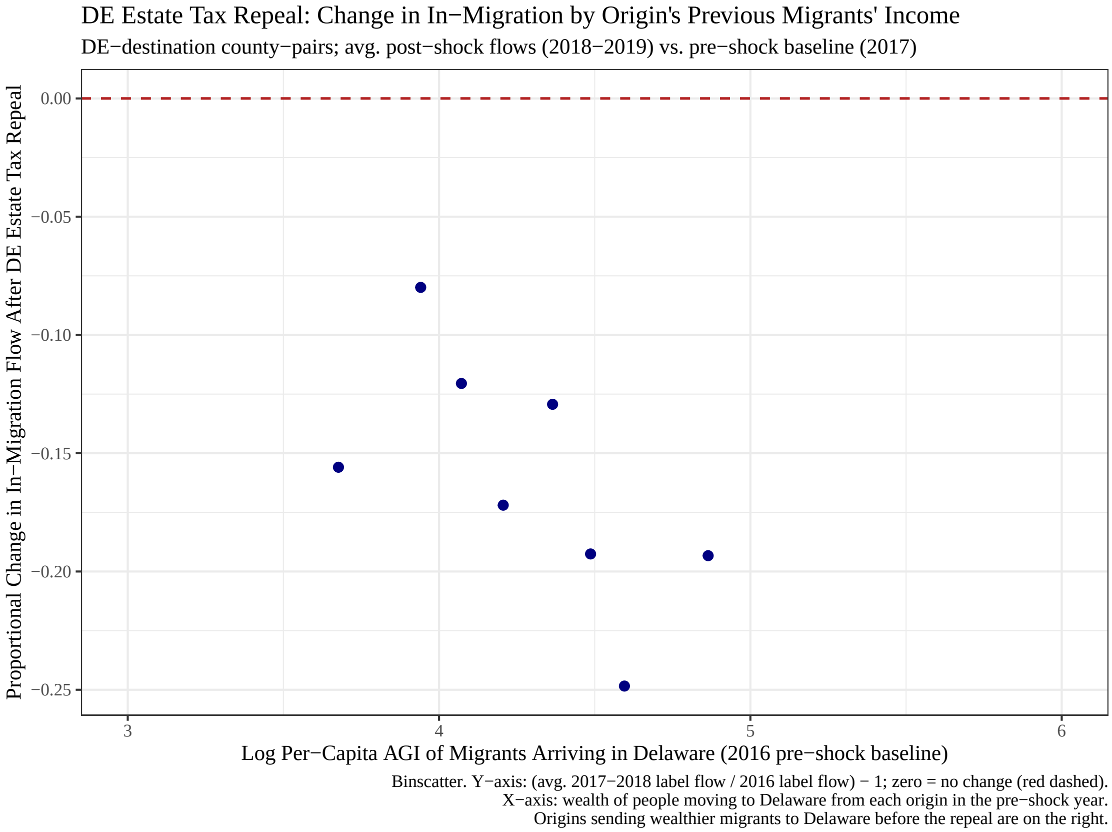


Figure 11: Delaware estate tax repeal: change in in-migration by origins' previous migrants' income.

From figure 11, it appears that counties from which migrants had higher average incomes had the largest decline in the number of people moving into Delaware after Delaware repealed their estate tax.

We estimate the effect on in-migration volume to Delaware:

$$log(n) = \alpha + \beta_1(Treat_c) + \beta_2(Post_t) + \beta_3(Treat \times Post)_{c,t} + \epsilon_{c,t}$$

where $Treat_c = 1$ if the destination county is in Delaware and $Post_t = 1$ for years 2017–2018 (moves during 2018–2019). We also estimate the income composition model:

$$log\left(\frac{AGI}{n}\right) = \alpha + \beta_1(Treat_c) + \beta_2(Post_t) + \beta_3(Treat \times Post)_{c,t} + \epsilon_{c,t}$$

Table 3: DE 2018 DiD Estimates. Standard errors in parentheses.

Coefficient	Volume: $\log(n)$	Income: $\log(AGI/n)$
β_1 (Treat)	-0.1148*** (0.0236)	0.2558*** (0.0116)
β_2 (Post)	0.2276*** (0.0035)	0.0876*** (0.0017)
β_3 (Treat $\times$ Post)	-0.0365 (0.0591)	0.0215 (0.0290)

Again, the Delaware regression results provide evidence against the notion that the estate tax repeal would produce a large increase in migration into Delaware. The volume model shows that Delaware destination flows were lower than other flows on average, while the post-period coefficient shows that migration volume rose across the board after 2017 in the control group. The interaction coefficient is negative but not statistically significant, meaning that we cannot conclude that Delaware received either more or fewer migrants compared to other destinations post-repeal. The income model, which excludes non-positive migrant per-capita AGI before logging, shows that migrants moving to Delaware had higher average AGI than migrants moving to other destinations; however, the interaction term is small, positive, and statistically insignificant. Overall, the estate tax repeal does not appear to have any relationship with an increase in either the number or average income of migrants moving to Delaware.

The figures below show mean log in-migration volume and mean log per-capita AGI over time for flows into Delaware counties (treated) versus flows into all other destinations (control). The dashed line marks the start of the post period.

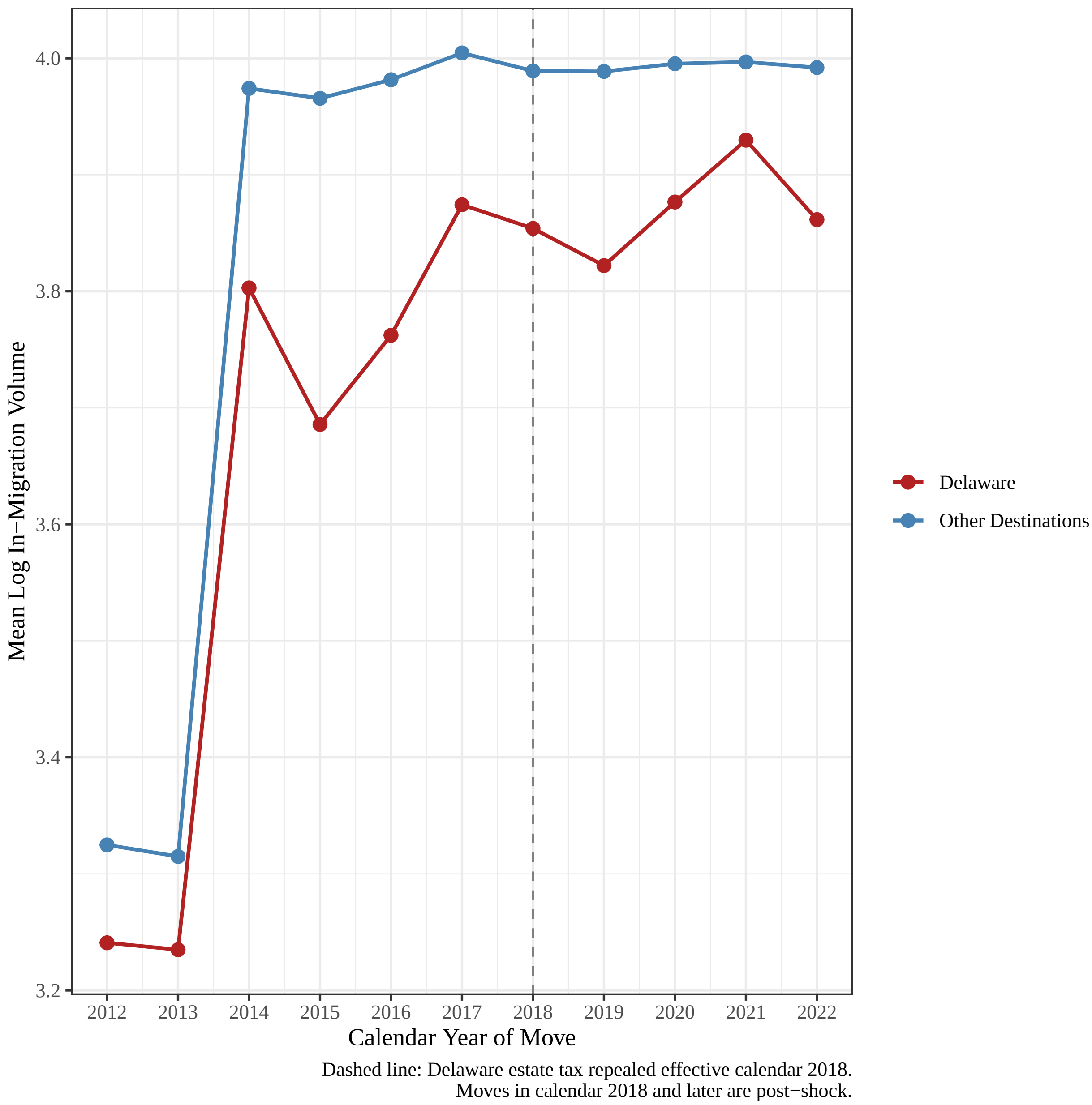


Figure 12: Mean log in-migration volume over time for Delaware destinations vs. other destinations.

The volume trends plot shows no clear post 2017 increase in Delaware in-migration relative to other destinations, suggesting that the tax repeal did not visibly change Delaware's migration volume trend.

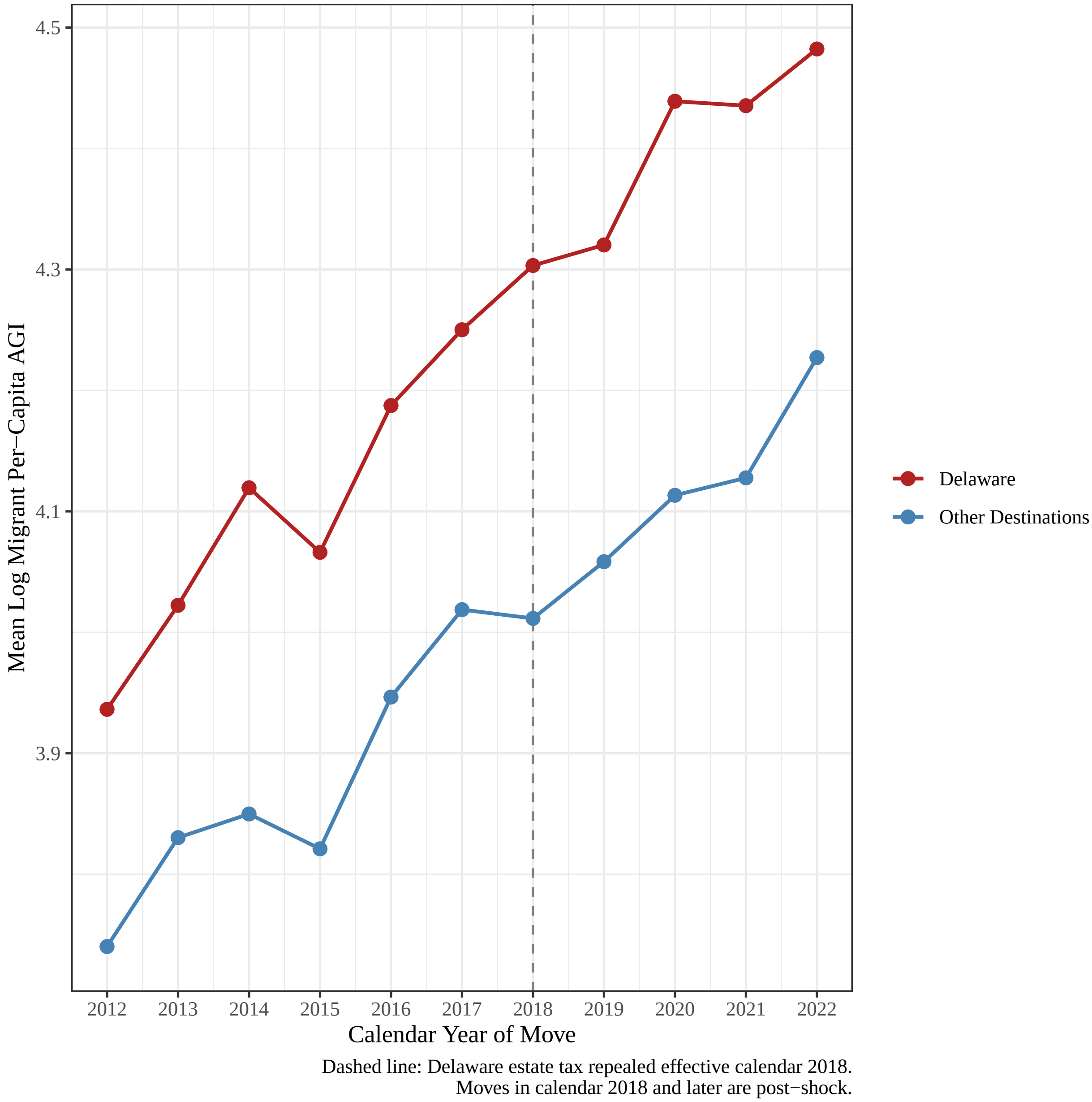


Figure 13: Mean log migrant per-capita AGI over time for Delaware destinations vs. other destinations.

The income trends plot shows that migrants moving into Delaware had relatively high per-capita AGI, but the post repeal pattern does not show a visible break from the control group's trend.

6 Exploring Variation in Tax Rates

To better understand the nature of these policy changes and contrast them to other sources of variation between counties in effective tax rates, we run the following TWFE:

$$log(n)_{p,t} = \beta_1 I(TaxShock)_{p,t-1} + \beta_2 log(\frac{Agi}{n})_{p,t-1} + \beta_3 I(TaxShock)_{p,t} * log(\frac{Agi}{n})_{p,t-1} + \gamma_{c1} + \delta_t + \epsilon_{p,t}$$

Where n is the number of households in a county pair-year observation. The indicator for tax shock is one if the receiving county experienced an increase to the tax rate paid by households with agi above $200k relative to a potential change in the same rate from the sending county from year t-2 to year t-1 beyond a certain value. The second term is the average income of people moving in year t-1 in the county-pair p. The third term is the interaction. The gamma fixed effect is of the county in year 1 of the move. Delta is the fixed effect for year 1 of the move.

These tax shocks are **not** exogenous. This regression answers the question, is there a relationship between the number of people moving between a county pair and the mean income of people who made the same move the previous year. Secondly, does this relationship change if high-income households in the receiving county happened to experience an increase in their taxes relative to the rate paid by households in the 'sending' county. Note that a relative tax increase in the receiving county is congruent to a relative tax decrease in the sending county, and these are different observations in the data (the county-pair for people moving from Gallatin to Missoula county is different from the county-pair for people moving from Missoula to Gallatin county).

Suppose the threshold for the first term is 5 percentage points. There is much variation between counties in tax rates, but for a county's high-income households to experience a 5 percentage point tax increase relative to another is likely a symptom of policy change, although it does not need to be. We test different thresholds, from 0 to 9%, and they are summarized in figure 14.

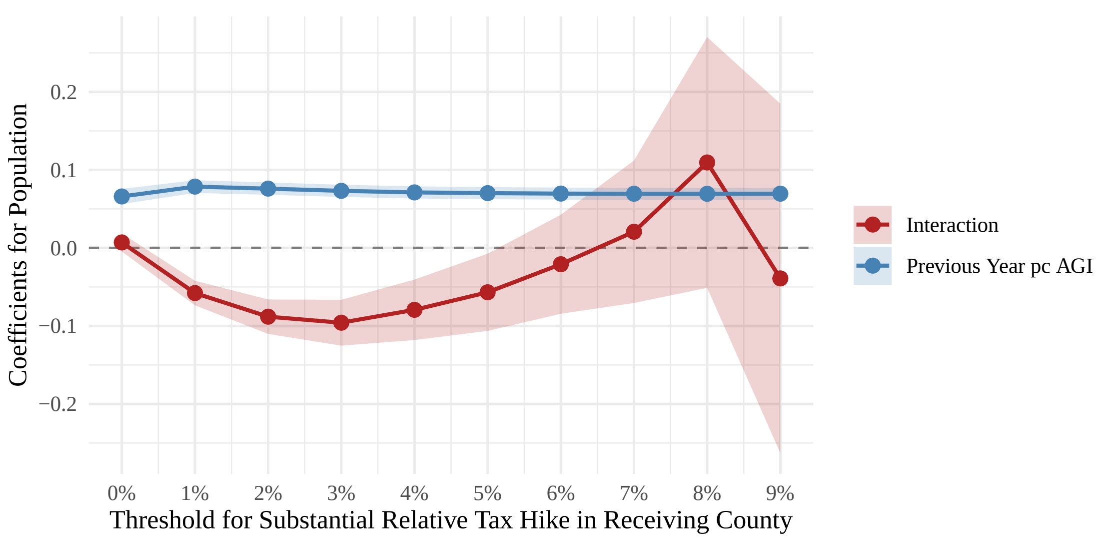


Figure 14: Corellations Between Migrating Population and Previous Migrants' Incomes, Distinguishing Between County-Pairs Experiencing Relative Realized Tax Changes

From the figure, every choice of relative tax change threshold in the model gives a very precise estimate, of 7%, for the relationship between mean income in the previous year and the migrating population in a pair of counties. **Interpretation:** A 1% increase in the average income of people migrating between a pair of counties is associated with a 0.07% increase in the number of people making the same move the next year. The relationship is very small.

When using 3 percentage points as the requirement to consider a relative tax increase in the receiving county 'substantial', a county experiencing the tax increase reduces the correlation between last year's migrants' incomes and the migrating population this year by a very precise 9.6 percentage points (from +7.3% to -2.3%).

The figure indicates that a county receiving a tax increase (to households above $200k in income) of 3 pp relative to another makes it less likely to attract high-income individuals, but the relationship is **reversed** if it experiences an even larger 8 pp, and reversed again if the change is 9 pp.

Next we consider the relationship between last year's migrants' incomes and this year's migrants' incomes. The model we use is:

$$log(\frac{AGI}{n})_{p,t} = \beta_1 I(TaxShock)_{p,t-1} + \beta_2 log(\frac{Agi}{n})_{p,t-1} + \beta_3 I(TaxShock)_{p,t} * log(\frac{Agi}{n})_{p,t-1} + \gamma_{c1} + \delta_t + \epsilon_{p,t}$$

Figure 15 summarizes results for different thresholds for 'substantial' tax increases.

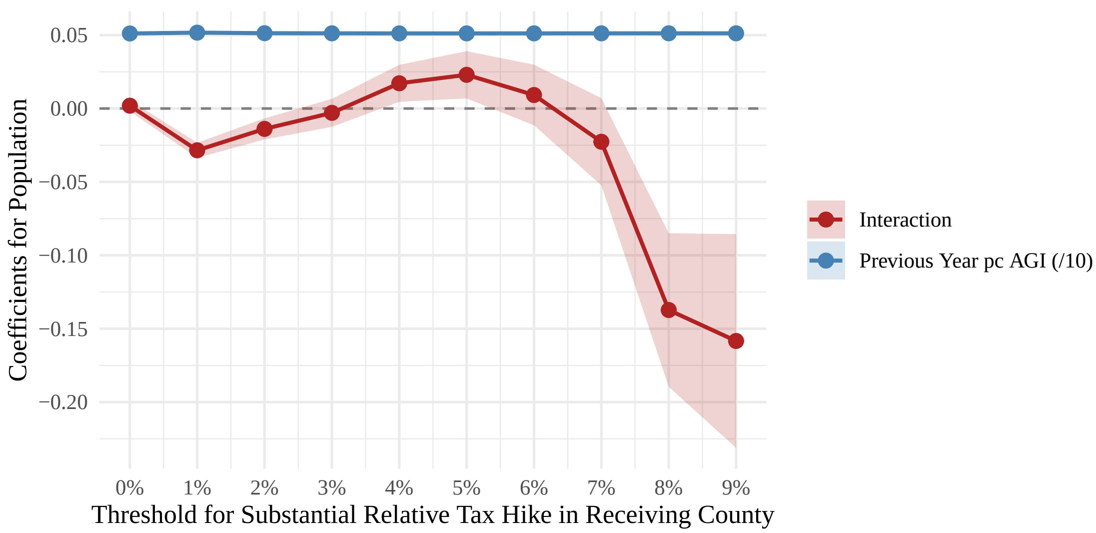


Figure 15: Corellations Between Migrants' Incomes and Previous Migrants' Incomes, Distinguishing Between County-Pairs Experiencing Relative Realized Tax Changes

Again, the coefficient on last year's migrants' incomes is massively postive and precise. It has been scaled down by an order of magnitude (the estimate is $\hat{\beta}_2 = 1/2$) to not obscure the more interesting picture of the interaction coefficients.

Households with AGI above $200k in a county experiencing an increase in their tax rate of 1 percentage point relative to another reduces the relationship between the average income of people migrating the previous year and the average incomes of people making the same move currently by 2.5 percentage points. Across all county pairs, a 1% increase in the average income of people moving between counties is associated with a 0.52% increase in the average income of people making the same move the next year. This 0.52% figure falls to 0.49% if the receiving county experienced a tax increase of 1 pp relative to the sending county. All the effects of the tax increase indicator are similarly small (although turning positive at 4 pp) until considering an 8 or 9 percentage point increase in the tax rate paid by high-income households in the receiving county relative to the sending county, at which point the affect of experiencing the tax increase decreases the relationship between last year's migrants' mean income and the current year's migrants' mean income from +0.51 to +0.38, which is still positive.

Oddly enough, the two figures looks nearly identical to their counterparts when considering tax hikes in the sending county instead of the receiving county. The relationship between last year's migrants' incomes and this year's migrating population turns negative when the county people are leaving experiences an increase in tax rates.

Something indicating a serious flaw of this exploration is that this TWFE model indicates a county that previously attracts higher-average-income people attracts fewer people if it experiences a 3 pp tax increase relative to where those people are moving from **and if it experiences a 3 pp tax decrease**.

Despite the apparent lack of signal in this TWFE model, it provides a way to proceed with a causal inference mission.

7 Causality Discusion

There is evidence of the NY tax hike encouraging people to leave NY. The increase in people leaving NY from before the tax change to afterward was 7% higher than the national average, and the incomes of these people increased by 7% more than the national average during the same period. This indicates the tax hike **did** push high-earners out of the state. HOWEVER, the tax hike happened at the same time as the pandemic, and it is almost certain that any increase in people leaving NY state is because the pandemic hit there particularly badly. It also is very likely, although not suggested by out data, that high-earning people are more mobile, so they were especially quick to flee pandemic-era NYC.

There is no evidence the NJ tax hike encouraged anybody to leave the state, and it had no effect on the distribution of the incomes of people leaving the state.

There is also no evidence the DE repeal of their estate tax encouraged any flow of people into the state. The increase from before the tax change to after in mean income of people changing counties was 2% greater among people moving into DE than otherwise, but this estimate is not statistically different from zero.

We figure the issue of using the TWFE model above is that the tax 'shocks' are not exogenous. The merits of it is that is gives a very highly powered model to examine the types of people who move between a pair of counties (described by their average incomes) and how the number (and average income) of people moving between these county-pairs changes when the two counties' relative tax levels change. To proceed further, we would use this to motivate more DDs. Some faults of our current DDs is that they do not actually measure the change in taxation in the nominally affected regions. We think the way to proceed toward causal inference is to try to identify what tax variation is exogenous, and regress the actual changes in tax rates against incomes of migrants.

Another problem with this approach to measuring the effect of tax changes is that our data doesn't distinguish between a marginal increase in people earning $100k relative to people earning $80 from a single family earning $1M deciding to move. These two changes in migration decisions are qualitatively different.

8 Reproducibility

This document was produced using **R** and **RMarkdown**. To reproduce it on any machine:

1. **Clone the repository** from GitHub.
2. **Open the project** by double-clicking `Data-Analysis-Project.Rproj` in RStudio. This sets the working directory to the project root automatically, do not open the `.Rmd` file directly from File Explorer/Finder, as paths will break.
3. **Knit the document**: open `Project Write-Ups/Stage_Two.Rmd` and click *Knit*, or run:

```
rmarkdown::render("Project Write-Ups/Stage_Two.Rmd")
```

4. **Packages are installed automatically** the first time you knit. No manual `install.packages()` calls are needed.
5. **The cleaned data file** (`Updated Working Files/Clean Data/cleanMigrationData.RData`) is committed to the repository. As long as this file is present, no scraping or data-cleaning scripts need to be re-run.

   Note on the map section (Section 5): The geographic figure downloads U.S. county shapefiles from the U.S. Census Bureau via the `tigris` package. This requires an internet connection on the first knit only; all subsequent knits use a local cache.

If `cleanMigrationData.RData` has been deleted, run the scripts in this order before knitting:

```
# Step 1 - only needed if Temp Data/ files were also deleted (requires internet)
source("Data Wrangling/Data Scraping.R")

# Step 2 - regenerates cleanMigrationData.RData from Temp Data/ files
source("Data Wrangling/Cleaning and Merging.R")
```

Session info at time of submission:

```
sessionInfo()
```

```
## R version 4.5.1 (2025-06-13 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.0
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: America/Denver
## tzcode source: internal
```

```
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] here_1.0.2      scales_1.4.0    sf_1.0-23       tigris_2.2.1
##  [5] fixest_0.13.2   binsreg_1.1     lubridate_1.9.4 forcats_1.0.1
##  [9] stringr_1.5.2   dplyr_1.1.4     purrr_1.2.1     readr_2.1.6
## [13] tidyr_1.3.1     tibble_3.3.0    ggplot2_4.0.0   tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] gtable_0.3.6        xfun_0.53           lattice_0.22-7
##  [4] tzdb_0.5.0          numDeriv_2016.8-1.1 vctrs_0.7.2
##  [7] tools_4.5.1         generics_0.1.4      sandwich_3.1-1
## [10] proxy_0.4-27        pkgconfig_2.0.3     Matrix_1.7-3
## [13] KernSmooth_2.23-26  RColorBrewer_1.1-3  S7_0.2.0
## [16] stringmagic_1.2.0   uuid_1.2-2          lifecycle_1.0.4
## [19] compiler_4.5.1      farver_2.1.2        MatrixModels_0.5-4
## [22] SparseM_1.84-2      quantreg_6.1        htmltools_0.5.8.1
## [25] class_7.3-23        yaml_2.3.10         Formula_1.2-5
## [28] pillar_1.11.1       MASS_7.3-65         classInt_0.4-11
## [31] nlme_3.1-168        tidyselect_1.2.1    digest_0.6.37
## [34] stringi_1.8.7       splines_4.5.1       rprojroot_2.1.1
## [37] fastmap_1.2.0       grid_4.5.1          cli_3.6.5
## [40] magrittr_2.0.4      survival_3.8-3      e1071_1.7-16
## [43] withr_3.0.2         dreamerr_1.5.0      rappdirs_0.3.3
## [46] timechange_0.3.0    rmarkdown_2.30      httr_1.4.7
## [49] matrixStats_1.5.0   hms_1.1.4           zoo_1.8-14
## [52] evaluate_1.0.5      knitr_1.50          rlang_1.1.7
## [55] Rcpp_1.1.1          glue_1.8.0          DBI_1.2.3
## [58] rstudioapi_0.18.0   R6_2.6.1            units_1.0-0
```